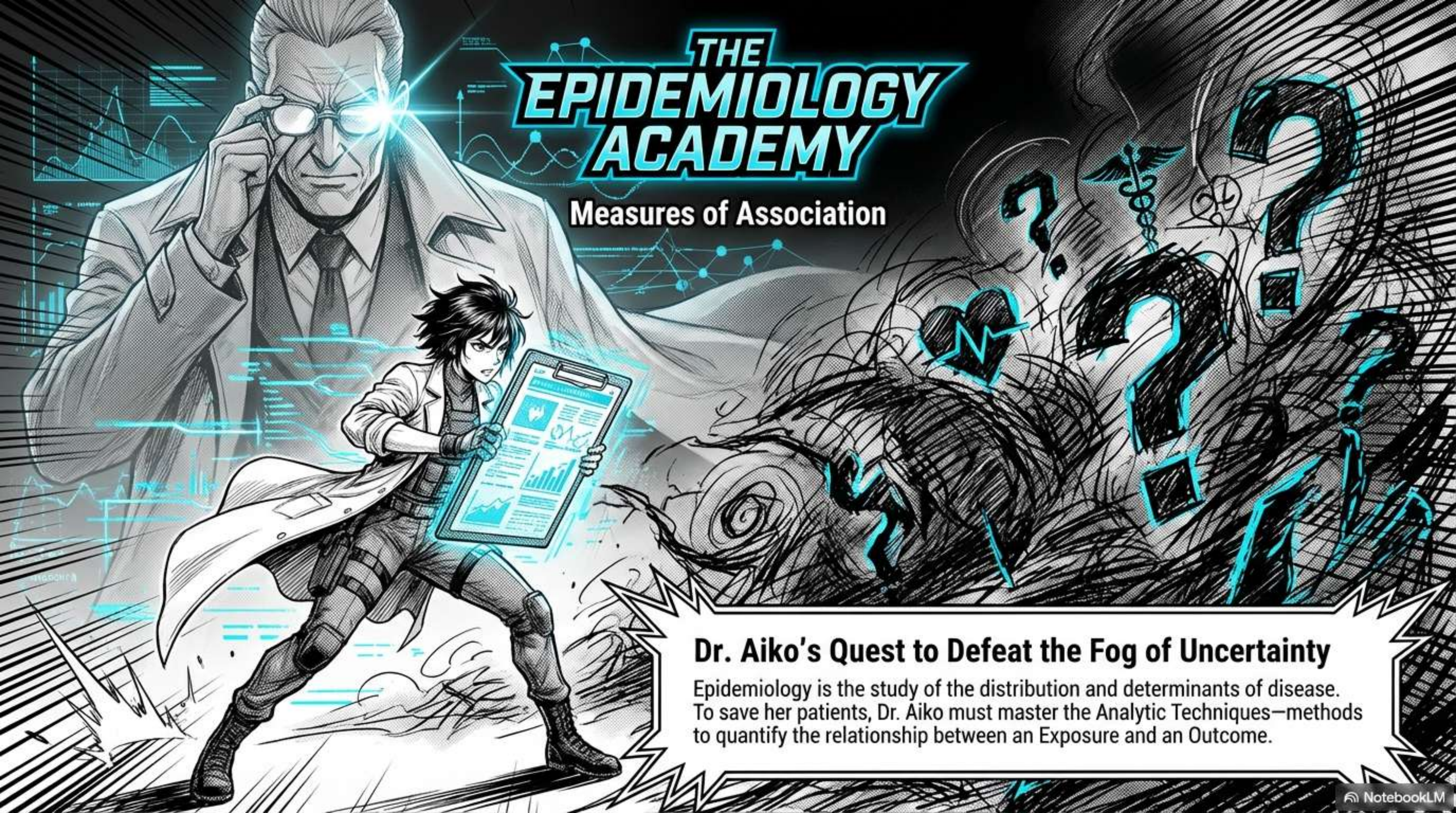




THE EPIDEMIOLOGY ACADEMY

Measures of Association

Dr. Aiko's Quest to Defeat the Fog of Uncertainty

Epidemiology is the study of the distribution and determinants of disease. To save her patients, Dr. Aiko must master the Analytic Techniques—methods to quantify the relationship between an Exposure and an Outcome.

Do not
fear the
chaos of
raw data,
Aiko.
Organize it.

Organize it.
This grid is
the shield
upon which
we map the
truth.

OUTCOME STATUS

EXPOSED

UNEXPOSED

EXPOSURE STATUS

YES
(DISEASE)

NO
(HEALTHY)

A

B

C

D

TECHNIQUE 1: THE FORWARD SLASH (Relative Risk)

For Cohort Studies

$$RR = \frac{(\text{Incidence in Exposed})}{(\text{Incidence in Unexposed})}$$

$$RR = \left[\frac{A}{A+B} \right] / \left[\frac{C}{C+D} \right]$$

**PROSPECTIVE
(Future)**

Source Data: Vitamin D Intake vs. Melanoma Risk

Comparison: Highest Quartile vs. Lowest

Result: $RR = 1.31$

Interpretation: The exposed group had 1.31 times the risk of the unexposed group.

TECHNIQUE 1: THE FORWARD SLASH (Relative Risk) (For Cohort Studies)

Careful!
Your point estimate
(1.31) suggests risk...
but look at the defense.

95% CONFIDENCE INTERVAL

RR = 1.31

0.94

1.82

1.0
(NULL VALUE)

The Confidence Interval
(0.94 – 1.82)
crosses the Null Value (1.0).

Result:
Not Statistically Significant.
The observed risk could be due to chance.

TECHNIQUE 2: THE BACKWARD GLANCE (Odds Ratio)

For Case-Control Studies



A Exposure in Cases	B Exposure in Controls
C No Exposure in Cases	D No Exposure in Controls

$$\text{Odds Ratio (OR)} = \frac{(\text{Odds of Exposure in Cases})}{(\text{Odds of Exposure in Controls})}$$

$$\text{OR} = \frac{A \times D}{B \times C}$$

Source Data: HPV Beta-Papillomavirus vs. Squamous Cell Carcinoma

Result: OR = 4.0 (95% CI: 1.3 – 12.0)

Interpretation: The CI does NOT cross 1. It is a direct hit! Cases were 4 times more likely to have the exposure than controls.

TRANSFORMATION SEQUENCE

OR

RARE
DISEASE
ASSUMPTION
($<10\%$)

RR

When the disease is rare, the cell counts A and C are tiny.
Therefore: $(A+B) \approx B$... and ... $(C+D) \approx D$. The Odds Ratio
gains the power to
approximate the Relative Risk.

CAUTION: If the disease is common ($>10\%$),
using OR will exaggerate the risk estimation!

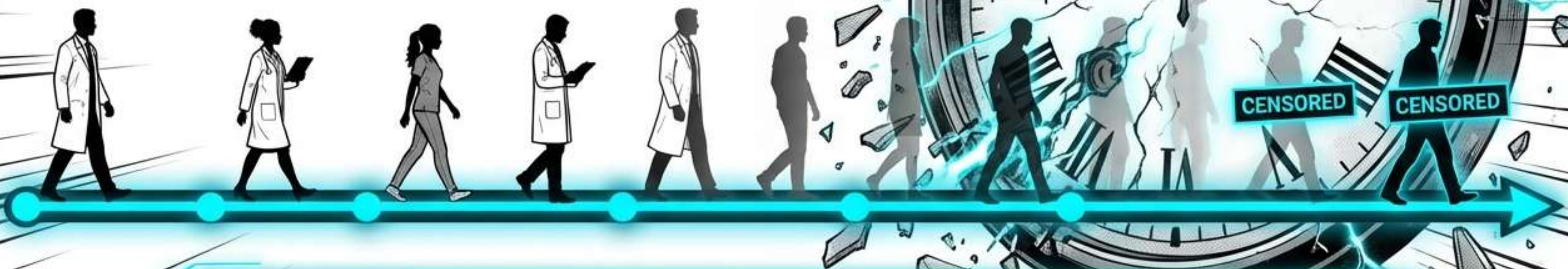


TECHNIQUE 3: THE TIME-STOPPER (Hazard Ratio)

For Survival Analysis

Hazard Ratio (HR): Measures the instantaneous rate of an event, not just the total count.

Handles "Right-Censoring": Participants who drop out or haven't had the event by the study's end.



Example: Systemic Immune Suppression vs. Merkel Cell Carcinoma Death

HR = 3.8 (95% CI: 2.2 – 6.4)

Interpretation: Immunosuppressed patients had 3.8x the rate of death compared to non-suppressed.

THE SUPPORT SQUAD: Correlation & Regression

Pearson (r)
Linear Strength

Spearman (rho)
Rank / Monotonic

Linear Regression (Beta Coefficient):

Source: Pain Level (0-10) vs. Sleep Disturbance
Beta = 0.21

Meaning: For every 1-point increase in pain,
sleep disturbance rises by 0.21 points.

CATEGORICAL DATA & ABSOLUTE IMPACT



Chi-Squared Test: Checks for independence between categories.

Example: Surgery Type vs. Clinician Level
 $P < 0.0001$ (Highly Significant difference in who performs which surgery)

Two tables are shown side-by-side. The first table is titled 'SURGERY TYPE' and the second is titled 'CLINICIAN LEVEL'. A blue arrow points from the first table to the second. Below the tables, the text $P < 0.0001$ is displayed in a large, bold, blue font.

SURGERY TYPE	

CLINICIAN LEVEL	

Number Needed to Treat (NNT)
Example: UV Light for Psoriasis.

$NNT = 16.7$

Translation: You must treat ~17 patients to see benefit in 1 patient.



~17 Patients

THE TRAP: BIAS

Bias is a systematic error. It pulls your estimate away from the truth, causing Overestimation or Underestimation.



THE BOSS: CONFOUNDING

The Shadow causes both! High Vitamin D looks like a risk (OR 1.85), but it's actually the Sun doing the damage.



SKIN
CANCER

CONFOUNDER
(Sun Exposure)

VITAMIN D

EXPOSURE
(Vitamin D)

OUTCOME
(Skin Cancer)

DEFENSIVE MANEUVER: ADJUSTMENT

By statistically adjusting for Sun Exposure, the shadow is banished. The true association is weak.

Other Defenses: Randomization, Matching, Stratification.

Unadjusted OR: 1.85
(Spurious High Risk)



Adjusted OR: 1.15
(The Truth)

MULTIVARIATE
ADJUSTMENT



FINAL ASSESSMENT: POWER & PRECISION

Do not worship the P-value (<0.05). It is arbitrary. Look at the Effect Size and the width of the Confidence Interval to judge clinical relevance.

POWER

Sample Size

Risk of False Negative

PRECISION

Wide CI
(Low Precision)
CI: [0.2, 2.0]

Narrow CI
(High Precision)
CI: [0.9, 1.1]

GRADUATION: THE MASTER SCROLL

Cohort Study



Relative Risk (RR)

Case-Control /
Cross-Sectional



Odds Ratio (OR)

Survival Analysis



Hazard Ratio (HR)

Continuous Linear Data



Pearson Correlation (r)

Categorical Comparison



Chi-Squared Test

**Sample
Size**

Risk of False
Negative

CI
(Confidence
Interval)
[1.5, 2.0]

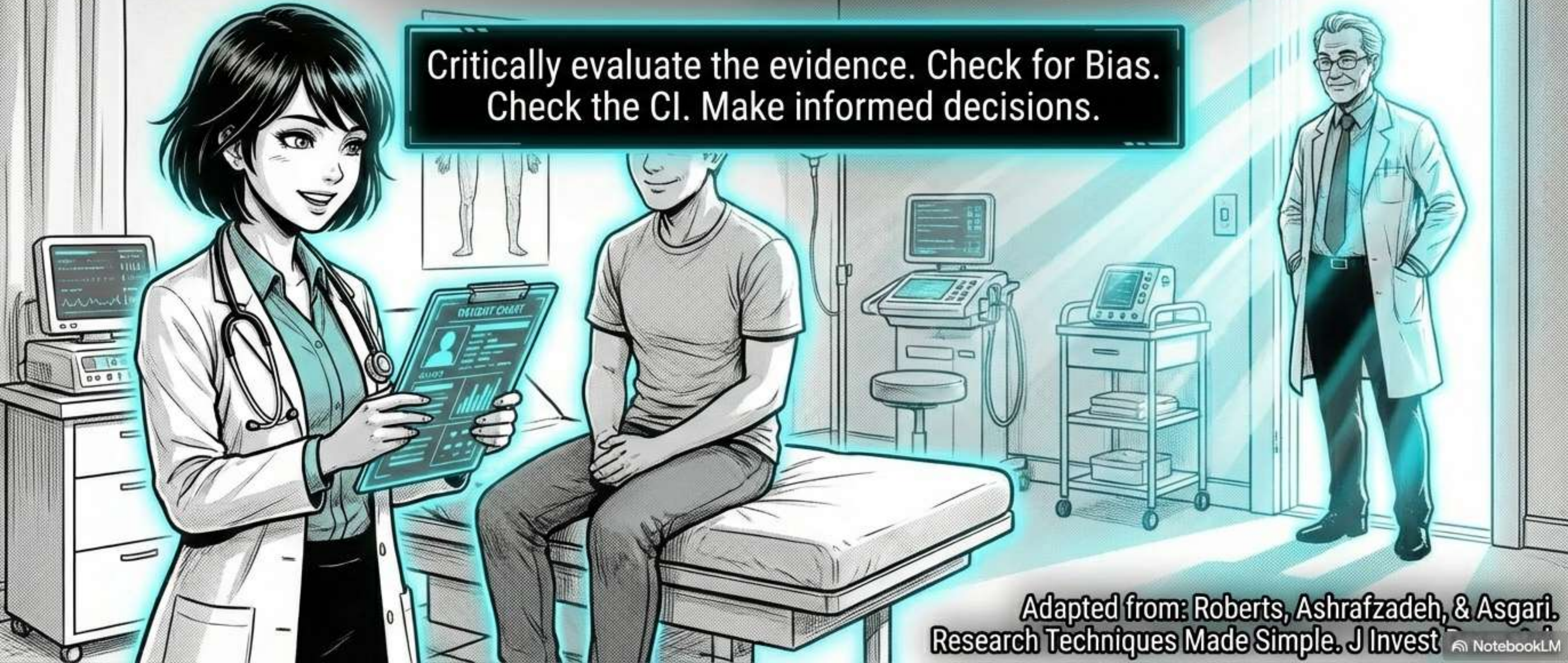
Low CI
(Confidence
Interval)
[1.1]

The measure you choose depends on the design you use.

EPILOGUE

By understanding Measures of Association, we bridge the divide between basic science and clinical practice.

Critically evaluate the evidence. Check for Bias.
Check the CI. Make informed decisions.



Adapted from: Roberts, Ashrafzadeh, & Asgari.
Research Techniques Made Simple. J Invest  NotebookLM